Recurrent Models for Trajectory Prediction (Long)

PaperLens Fixtures

1. Section 1

In air traffic management (ATM), trajectory prediction is a core function. Controllers rely on accurate forecasts of aircraft positions, and modern decision support tools increasingly combine physical models of the atmosphere with data-driven predictors such as LSTMs.

We train a stacked LSTM on surveillance data and compare it with a Transformer encoder that uses multi-head self-attention. The Long Short-Term Memory baseline is surprisingly competitive on short horizons, while the transformer architecture performs better beyond ten minutes.

Each training sequence contains position, altitude and ground speed sampled every five seconds. We normalise all features, remove outliers caused by radar artefacts, and split the dataset chronologically so that the evaluation period never overlaps with the training period.

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Ablation experiments show that removing the forget gate degrades accuracy by a large margin, which confirms that the gating mechanism of Long Short-Term Memory is essential. Bidirectional LSTM variants did not help, because future observations are unavailable at prediction time.

Finally, we discuss limitations. The model ignores weather information and the intentions of pilots, and it is evaluated on a single region of airspace. Extending the approach to other regions, and combining it with physics-based models of the atmosphere, is left for future work.

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